Build React + TypeScript apps with Vite

Fast dev server • Instant HMR • Rollup-powered production builds

TL;DR: Vite serves source as native ESM in dev and bundles for production.

```
npm create vite@latest my-app -- --template react-ts
cd my-app
npm install
npm run dev
```

Why Vite works so well for React + TypeScript

Speed in dev, sensible defaults, and a flexible plugin ecosystem

Fast dev server

- Enhances native ESM for a smooth dev experience
- Pre-bundles dependencies for faster startup
- Instant page loads as modules are requested

Great feedback loop

- Extremely fast Hot Module Replacement (HMR)
- First-party React Fast Refresh integration
- TypeScript transpile

Production-ready builds

- Uses Rollup for optimized production bundles
- Tree-shaking + code splitting out of the box
- Configurable via vite.config.ts + plugins

Scaffold a React + TypeScript project

The official “react-ts” template gives you TSX, Vite scripts, and a clean starting point

Quick start

- 1) Create the project
- 2) Install dependencies
- 3) Run the dev server
- 4) Build + preview locally

What you get

- src/ with TSX entry + components
- vite.config.ts for build + plugins
- Scripts: dev, build, preview
- Easy path to linting/testing via plugins

```
# scaffold
npm create vite@latest my-app -- --template react-ts

# run
cd my-app
npm install
npm run dev

# build + test
npm run build
npm run preview
```

Developer workflow: fast refresh with TSX

Edit → save → Vite updates the browser without a full reload

Edit TSX

Change a component
(or a hook / style)

Vite dev server

Transforms modules
+ serves native ESM

Browser update

HMR + React Fast Refresh
keeps component state

Build, preview & customize

Production bundles, environment variables, and vite.config.ts

Key commands

```
npm run build   # outputs dist/
npm run preview # serve dist locally
```

Environment variables

Vite exposes client-safe vars under import.meta.env. Only variables prefixed with VITE_ are exposed to your client code.

```
// .env
VITE_API_URL=https://api.example.com

// in code
fetch(import.meta.env.VITE_API_URL)
```

vite.config.ts pattern

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

export default defineConfig({
  plugins: [react()],
  resolve: {
    alias: {
      '@': '/src',
    },
  },
})
```

Reminder: vite preview is for local smoke tests—don't use it as a production server.